Inward displacement pays for conditional noise

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A centered noise cannot have large variance and a small upper endpoint without a negative third moment. Keeping that third moment in the gap budget gives a stronger noise cost. This argument uses the actual central gap; it needs neither the outer support strips nor a count comparison. It reduces the acquired global noise ceiling from .616 to .36, and to .32 on the left range. It also confines the part below $p = 1/3$ to the indices $42 \le n \le 168$.

1 The common-law geometric inequality

Let X be centered, with variance one and compact support, with an actual central gap $(-A_-, A_+)$ and no atom at zero. Let n be a positive integer. Write

$$p = \mathbb{P}(X > 0), \quad q = 1 - p, \quad v = pq, \quad m = \mathbb{E}X_+ = \mathbb{E}X_-, \quad \ell = m/v.$$

Use its actual sign $B = \mathbf{1}_{X>0}$ to write $X = \ell(B + U - p)$, so $\mathbb{E}(U \mid B) = 0$. Put

$$a_0 = p - A_-/\ell, \qquad a_1 = q - A_+/\ell, \qquad d = a_0 + a_1, \quad s_j = \mathbb{E}(U^2 \mid B = j),$$

$$s = qs_0 + ps_1, \quad \lambda = ns, \quad \lambda_0 = nqs_0, \quad \lambda_1 = nps_1.$$

Thus $a_j \ge 0$, $U_0 \le a_0$, $U_1 \ge -a_1$, $\lambda_0 + \lambda_1 = \lambda$, and $\ell = (v + s)^{-1/2}$. For $\sigma = \sqrt{10}$ define

$$Q(p) = (\sigma + 1)p^2 + (\sigma - 1)q^2 - 6pq = (6 + 2\sigma)(p - p_*)^2, \qquad p_* = (4 - \sigma)/2,$$

$$D = (3 + \sigma)\rho - \gamma - 3J, \quad \rho = \mathbb{E}|X|^3, \quad \gamma = \mathbb{E}X^3, \quad J = \sum_{(a,b)} \tau_{a,b}(b - a)^2.$$

The sum is over the bounded support gaps and $\tau(t) = \mathbb{E}[X\mathbf{1}_{X \ge t}]$ is constant on each gap. There are at most countably many gaps. Their nonnegative contributions are interpreted as a monotone sum; the integral bounds below apply to each finite subcollection and then to the full sum by monotone convergence. Compact support makes the displayed moment bounds finite. All quantities in this note refer to one actual law.

Lemma 1 (One-sided cubic cost)**.** *If Z is centered, $Z \le a$ almost surely, $a > 0$, and $t = \mathbb{E}Z^2$, then*

$$\mathbb{E}Z^3 \le at - t^2/a.$$

If $a = 0$, centering forces $Z = 0$ almost surely.

Proof. Expand the nonnegative expectation $\mathbb{E}[(a - Z)(Z + t/a)^2] = at - \mathbb{E}Z^3 - t^2/a$. No lower support endpoint is used. Apply this also to $-U_1$. □

Fix $0 \le \bar{d} < 1$ with $d \le \bar{d}$, and define

$$Y_0(p) = 3(\sigma + 3)p - (\sigma + 7)\bar{d} + 2\sqrt{6(1 - \bar{d})(\sigma + 1)p},$$

$$Y_1(p) = 3(\sigma + 1)q - (\sigma + 5)\bar{d} + 2\sqrt{6(1 - \bar{d})(\sigma - 1)q}.$$

Theorem 2 (A probability–noise–allocation budget)**.** *Under the preceding hypotheses,*

$$nD \geq \frac{nvQ(p) + \lambda_0 Y_0(p) + \lambda_1 Y_1(p)}{(v + \lambda/n)^{3/2}}. \tag{1}$$

In particular the numerator is at least $nvQ(p) + \lambda \min(Y_0(p), Y_1(p))$.

Proof. We retain the two residual noise measures separately in the direct gap budget. For example the positive outer gaps satisfy

$$J_o^+ \leq 2 \int_{A_+}^{\infty} (t - A_+)\tau(t)\, dt = \mathbb{E}[X_+(X_+ - A_+)^2],$$

with the reflected expression on the negative side. Adding the actual central gap gives the exact lower-budget expansion used in v10:

$$\begin{aligned} nD(v+s)^{3/2} \geq\ & nvQ + 6nv\{d(1-d) + a_0 a_1\} \\ & + 3(\sigma+3)p\lambda_0 + 3(\sigma+1)q\lambda_1 - 6a_0\lambda_0 - 6a_1\lambda_1 \\ & - (\sigma+1)nq\mathbb{E}U_0^3 + (\sigma-1)np\mathbb{E}U_1^3. \end{aligned}$$

This is the expansion of the common geometric cost before replacing a conditional cubic by its endpoint bound; see `mathematics/simplification-v10/2026-10-28/geometry.tex`, the proof of the common geometric cost. The integration includes endpoint atoms; individual integration endpoints have Lebesgue measure zero.

Lemma 1 improves the last line, giving

$$\begin{aligned} nD(v+s)^{3/2} \geq\ & nvQ + 6nv\{d(1-d) + a_0 a_1\} \\ & + \{3(\sigma+3)p - (\sigma+7)a_0\}\lambda_0 + \{3(\sigma+1)q - (\sigma+5)a_1\}\lambda_1 \\ & + \frac{(\sigma+1)\lambda_0^2}{nqa_0} + \frac{(\sigma-1)\lambda_1^2}{npa_1}. \end{aligned}$$

A term with $a_j = 0$ has $\lambda_j = 0$ and is simply absent; there is no division by zero. Bound $a_j \leq \bar{d}$ in the two negative linear terms and use $d(1-d) + a_0 a_1 \geq (1-\bar{d})(a_0 + a_1)$. For each j, apply the arithmetic–geometric mean inequality to its central-displacement and reciprocal-cubic costs. For example

$$6nv(1-\bar{d})a_0 + \frac{(\sigma+1)\lambda_0^2}{nqa_0} \geq 2\lambda_0 \sqrt{6(1-\bar{d})(\sigma+1)p}.$$

The other group is identical with p, q and $\sigma+1, \sigma-1$ interchanged. This proves (1). □

The allocation λ_1/λ has not been optimized independently of the law. The full inequality may be retained as a shared resource; replacing its affine allocation term by its smaller endpoint is only a conservative simplification. No contact, convolution or predecessor assumption was used in Theorem 2.

2 Consequences for the acquired domain

The following are existing global geometric outputs of the selected coarse entry, not conclusions of a count calculation:

$$n \geq 42, \quad p \in [1/4, 27/50], \quad d \leq 17/250, \quad 0 \leq \lambda \leq 77/125, \quad nD \leq B_* := 32.20665378. \tag{2}$$

The strict $p > 1/4$, $d < 17/250$ and $\lambda < 77/125$ previously proved may safely be weakened to these closed premises. Only the incoming coarse entry and direct gap geometry are needed; the .6/.7 support strips are not needed for the conclusions below.

Theorem 3. *Under* (2)*, one has the following strict noise ceilings:*

p at most	3/10	1/3	2/5	27/50
λ less than	1/6	1/4	8/25	9/25

Moreover the following integer-index implications hold:

p at most	.30	1/3	.35	.375	.39	.40
n at most	84	168	262	657	1530	3598.

Proof. Set $H_j = Y_j/v$. On the stated domain, $v \geq 3/16$, $0 \leq Q \leq 9/25$ and $H_j > 30$. These elementary bounds are checked with rational square comparisons. For fixed p, put

$$F_j(n,\lambda) = \frac{nQ + H_j\lambda}{\sqrt{v}\{1+\lambda/(nv)\}^{3/2}}.$$

Theorem 2 gives $nD \geq \min_j F_j$. Writing $z = \lambda/n \leq 11/750$, the sign of its derivative in λ is that of

$$H_j\{1 - z/(2v)\} - 3Q/(2v) \geq 30\{1 - (11/750)/(2(3/16))\} - 3(9/25)/(2(3/16)) > 0.$$

For fixed positive λ, F_j also increases in n: apart from a positive factor it is $(Q/z + H_j)/(1 + z/v)^{3/2}$, which decreases in z. Thus to exclude $\lambda \geq L$ it suffices to check $F_0(42, L) > B_*$ and $F_1(42, L) > B_*$ on the relevant probability interval. This is an algebraic monotonicity argument, not a perturbation or limit of actual laws.

Here is a small exact certificate for these eight continuous inequalities. Use

$$s_- = 3.162277660 < \sigma < s_+ = 3.162277661.$$

In each Y_j replace its positive coefficients by s_- and its negative coefficient by s_+. Replace its square-root term by the chord joining rational lower bounds at the two interval endpoints. Concavity of the square root puts this affine chord below the square root on the entire interval. Call the resulting affine polynomial $y_j(p)$, and let Q_- denote Q with σ replaced by s_-. For $N = 42$ form

$$R_j(p) = NvQ_-(p) + Ly_j(p), \qquad P_j(p) = N^3R_j(p)^2 - B_*^2(Nv + L)^3.$$

Positive R_j and P_j prove the desired inequality without a sign ambiguity from squaring. The polynomial degrees are four and eight. For the four intervals and ceilings in the first table, all degree-four Bernstein coefficients are positive on the whole interval. The degree-eight coefficients are also positive on the whole interval, except for the global $j = 1$ polynomial, for which the four equal closed subintervals suffice. There are 139 strictly positive coefficients in total, on eleven closed polynomial intervals. The script constructs the polynomials from these formulas, rather than loading fitted coefficients. It also verifies each square-root enclosure before using it. This proves the first table.

Monotonicity at $\lambda = 0$ gives $D \geq Q(p)/\sqrt{pq}$. On $[1/4, 2/5]$ this ratio decreases: Q is positive and decreasing, while pq is increasing. For each upper endpoint b and index N_b in the second table, the exact positive-base comparison is

$$(N_b + 1)^2Q_-(b)^2 > B_*^2 b(1 - b).$$

It excludes $n \geq N_b + 1$ when $p \leq b$, proving the second table. □

The same inequality gives explicit near-critical localization:

$$|p - p_*| \leq \sqrt{\frac{B_*}{2(6 + 2\sqrt{10})n}} < \frac{23}{20\sqrt{n}}.$$

In particular no acquired witness with $n \geq 169$ lies below 1/3, and no acquired witness with $n \geq 3599$ lies at or below .4. These remove whole probability/index regions before count comparison; they do not certify the remaining finite indices.

3 The Bernoulli gap: exactly the available scope

Schulz's original thesis, *The optimal Berry–Esseen constant in the binomial case* (2016), Theorem 1 on printed page 1, supplies the strict finite-n bound at C_E for every $0 < p < 1$. Its sharper formula is stated only for $p \in [1/3, 2/3]$; see the original thesis. For $1/3 \le p \le 1/2$, that formula yields the weighted reserve

$$C_E\rho_B - \sqrt{n}\sup_x |F_B(x) - \Phi((x-np)/\sqrt{nv})| > \frac{Q(p)}{6\sqrt{2\pi v}} = \frac{2C_E(p-p_*)^2}{\sqrt{v}}.$$

This is an exact algebraic identity, not a new extension of the external theorem. Below $1/3$ we retain only strictness. The geometric index bound now confines that unresolved quantitative-gap range to $42 \le n \le 168$. No Bernoulli-gap premise is used in the new noise ceiling.

The thesis gives a warning example on printed pages 26–27: $n = 217$, $p = 77/2604$, evaluated at $k = 6$. Our exact rational normal enclosure independently confirms that the sharper formula fails there. This lies outside the acquired probability interval and does not refute its possible extension to $[1/4, 1/3)$; it rules out an unrestricted extrapolation.

4 Checks, an obstruction, and the remaining work

The scalar checker uses exact rational arithmetic for the 139 coefficients and all denominator, monotonicity and integer-index gates. A separate actual-law check computes every support gap and its weight, retaining threshold atoms, rather than treating conditional means as support points. It includes zero conditional noise, both variance allocations, symmetric and skew Bernoulli limits, tiny positive defects, and inward cubic equality cases. Each positive-noise law is also checked after reflection, with its actual sign probability and conditional variance allocations exchanged. These examples test the formulas; they do not prove the continuous domain by sampling.

There is a concrete limit to the reduction. Take $n = 42$, $p = 3/10$, $U_1 = 0$, and let the centered law U_0 be supported on $\{-3/5, 5/882\}$. Then $\lambda = 1/10$, $d = 5/882$, and the standardized law obeys the global support and geometric bounds, including $nD < 27 < 32.20665378$. It lies below $1/3$ and above the completed left count range .01. Thus these necessary geometric data alone do not remove every remaining left case. No contact or nonnegative-excess premise is asserted for this example.

The new bounds make all count work at $\lambda \ge .36$ unnecessary for an acquired witness, and all left count work at $\lambda \ge .32$ unnecessary. Smaller left probability ranges have the stronger ceilings in Theorem 3. Their derivation requires no outer-strip or heat/ray calculation. The coarse entry, actual gap, original geometric bootstrap and their contact/predecessor ancestry remain. Count comparison below the new ceilings, signed/reflected acquisition, and the independent finite base through 41 remain open obligations. No previous theorem, certificate or manuscript source is changed.